

Predicting Occupational Employment Decline

Kim Cates

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1. Introduction and Motivation

Long-horizon occupational forecasts are a central input to workforce policy, career counseling, and educational investment. The U.S. Bureau of Labor Statistics (BLS) publishes Employment Projections every two years; the 2024–2034 release covers 832 detailed occupations and projects that 29.3% of them (244 occupations) will see absolute employment decline over the ten-year window. The arrival of generative AI has raised the question of which industries are at risk of losing jobs.

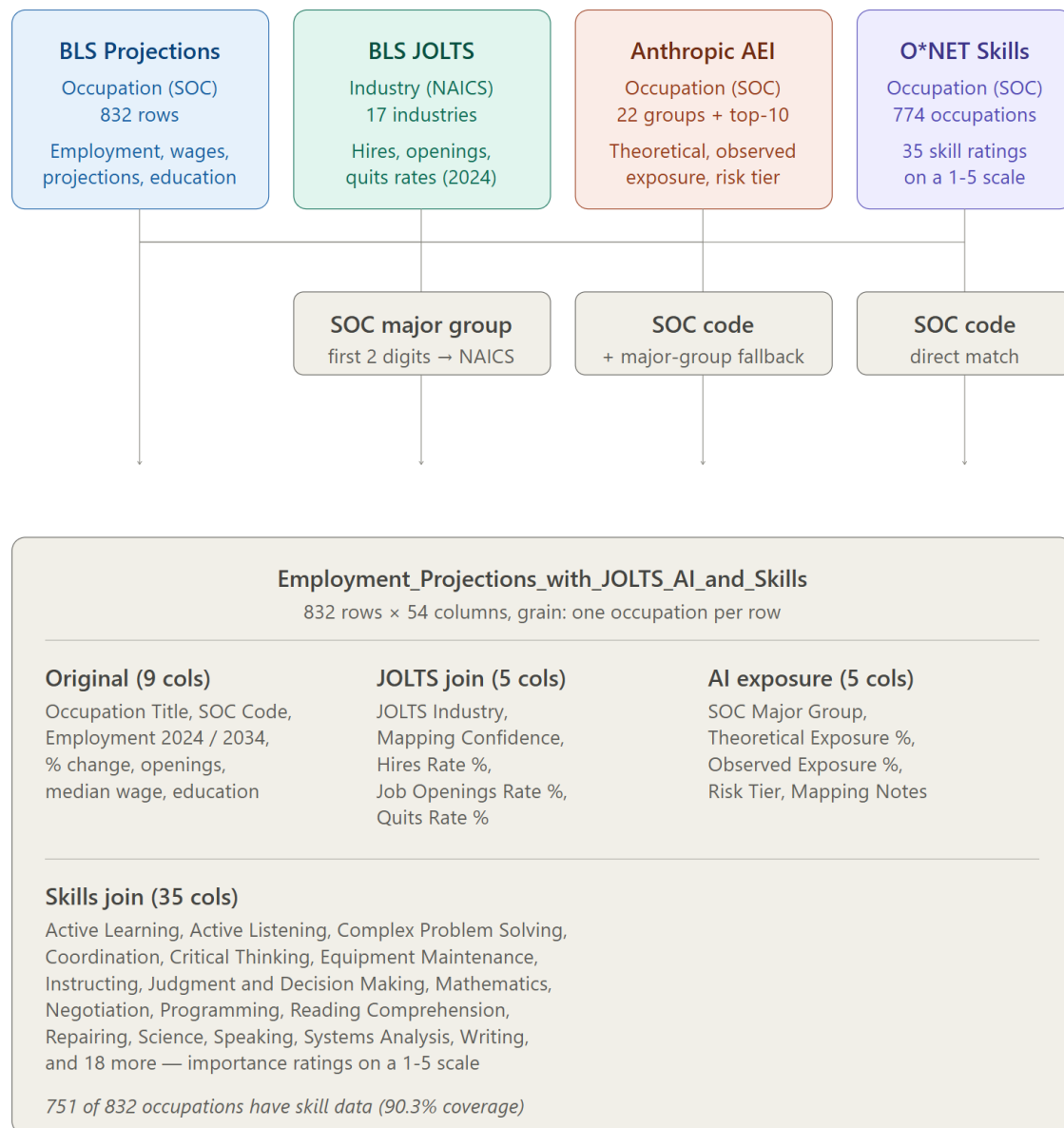
This project addresses which occupations are most at risk of projected employment decline and further investigates which skills and exposure patterns are associated with these declines. The analysis combines four data sources (BLS projections, JOLTS labor-market flows, the Anthropic Economic Index, and O*NET skill ratings) and finds that the skills most associated with projected decline are not the ones most exposed to AI.

2. Dataset Description

The working dataset is a single CSV with one row per occupation ($n = 832$), assembled from four sources by joining on the 6-digit Standard Occupational Classification (SOC) code:

- BLS Employment Projections, 2024–2034 release — employment levels for 2024 and 2034, projected percent change, annual openings, median wage, and typical entry-level education.
- BLS JOLTS (Job Openings and Labor Turnover Survey) — 2024 annual averages for hires rate, job openings rate, and quits rate, mapped from each occupation's SOC major group to the nearest NAICS industry through a hand-built crosswalk.
- Anthropic Economic Index (March 2026) — theoretical AI exposure (Eloundou β , share of tasks an LLM could plausibly perform) and observed exposure (share of tasks where Claude is empirically used), at the SOC major-group level with per-occupation overrides for the ten most-exposed roles.

- O*NET database version 30.2 (Skills file) — importance ratings on 35 standardized skills (1–5 scale), pivoted from long to wide format and joined to 751 of 832 occupations (90.3% coverage). The 81 unmatched occupations are mostly BLS "all other" catch-all codes that O*NET does not profile separately.



Three preprocessing steps were required. First, the Median Annual Wage, Employment 2024, and Employment 2034 columns contained thousand-separator commas and "N/A" literals; these were coerced to proper numeric types. Second, the BLS Occupation Title

column concatenates each canonical title with a long alias list; this was split to produce a clean title column. Third, the wage column had 6 missing values (occupations not separately surveyed); these were median-imputed for modeling. The O*NET importance ratings, log-transformed wage, and log-transformed 2024 employment were the final engineered features used in modeling, alongside ordinal-encoded education level and the AI exposure measures.

One challenge worth noting: the AI Displacement Risk Tier (categorical) was originally constructed from the AI Theoretical Exposure values, leading to a 0.97 correlation and variance inflation factors above 30 when both were included in regression. Risk Tier was dropped from the final model; the AUC was essentially unchanged, confirming the redundancy.

3. Research Questions

The analysis addresses three connected questions:

- Q1. What is the likelihood that a given occupation will experience employment decline between 2024 and 2034? How does the predicted probability behave across different probability cut-off thresholds?
- Q2. Which occupational characteristics (education, wage, employment size, AI exposure, industry churn rates) are most associated with projected decline versus growth?
- Q3. Which O*NET skills are most associated with projected decline, and separately with AI exposure? Do combinations of skills (extracted via PCA) improve predictive accuracy beyond the baseline features?

Disclosure: the dataset contains projected employment at the occupation level, not worker-level unemployment data. We therefore model the probability of occupation-level decline as a proxy for labor-market risk, and refer to it as $P(\text{Decline})$ throughout to avoid conflating it with individual unemployment probability.

4. Methods and Approaches

Two complementary supervised-learning models address Q1 and Q2. A logistic regression predicts the binary target Will Decline (1 if projected 2034 employment is below 2024 employment, else 0). A linear regression predicts the continuous Employment Percent Change, 2024–2034. Reporting both lets us check whether patterns are robust to the binarization. All features were standardized (z-scored) so coefficients are directly comparable; performance was evaluated with 5-fold cross-validation using AUC for the logistic model and R^2 for the linear model.

For Q3, three exploratory techniques were applied to the O*NET skill matrix. First, univariate Pearson correlations between each of the 35 skills and the decline target, separately for AI Observed Exposure. Plotting these two correlations against each other (Section 5.4) identifies a "displacement quadrant" or skills associated with both high AI exposure and decline. Second, Principal Component Analysis (PCA) on the 35 standardized skill ratings extracts a small number of orthogonal combined-skill factors. Third, a three-way model comparison was run on the 751-occupation subset with skill data: a base model using only the eight engineered features, a base+skills model adding all 35 skill ratings, and a base+PCs model adding the first four principal components. This

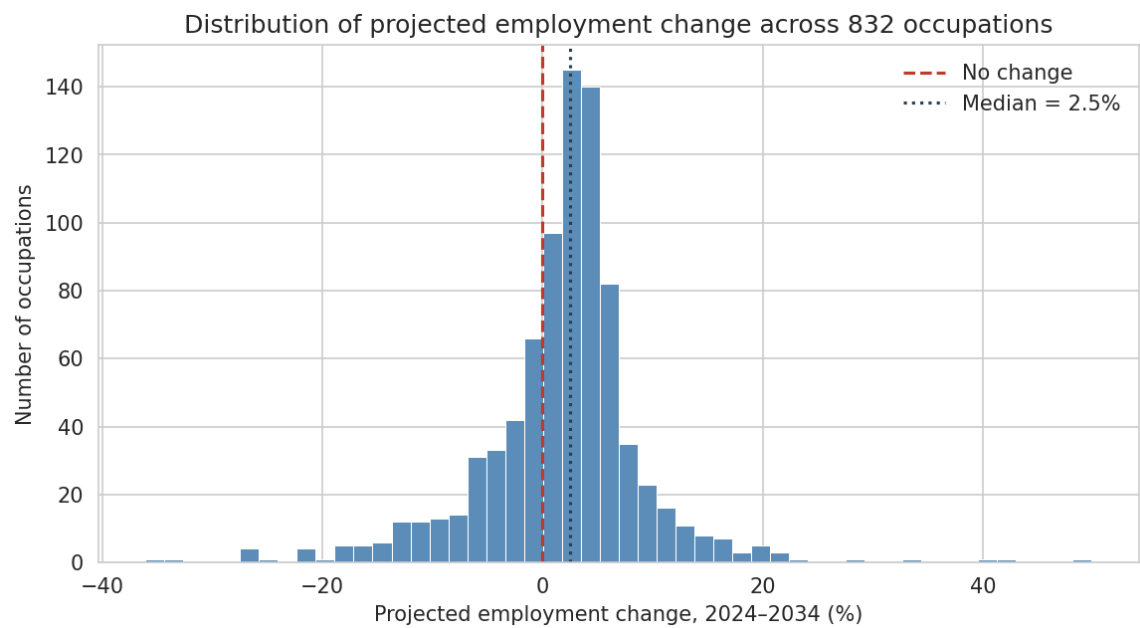
isolates whether skills carry predictive information not already captured by the base features.

Logistic regression was selected over more flexible classifiers (random forest, gradient boosting) for interpretability. Since there were 832 occupations and ~10 features, the marginal predictive gain from a nonlinear model is small, and standardized coefficients with odds ratios allow stakeholders to read the model directly. Variance inflation factors were checked before fitting (Section 4.2) to ensure features were not collinear, with Risk Tier dropped as described above.

5. Results and Analysis

a. Target distribution and base rates

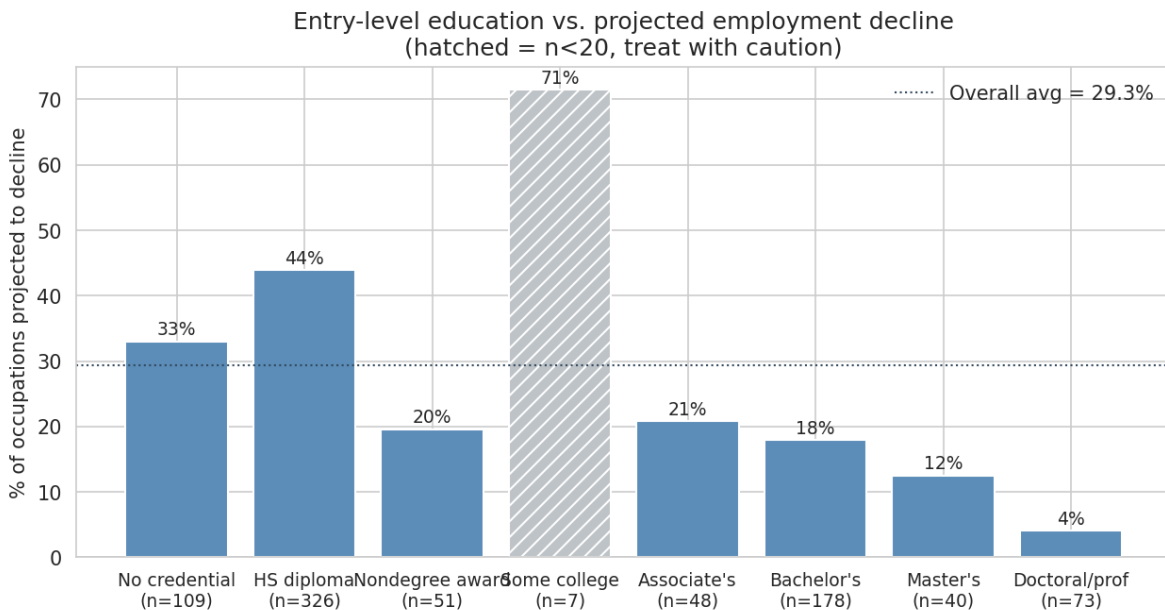
Of 832 occupations, 244 (29.3%) are projected to decline by 2034. The percent-change distribution is approximately normal, with a median of +2.5% and a long left tail of contracting occupations (Figure 1).



b. Decline rates by AI risk tier and education

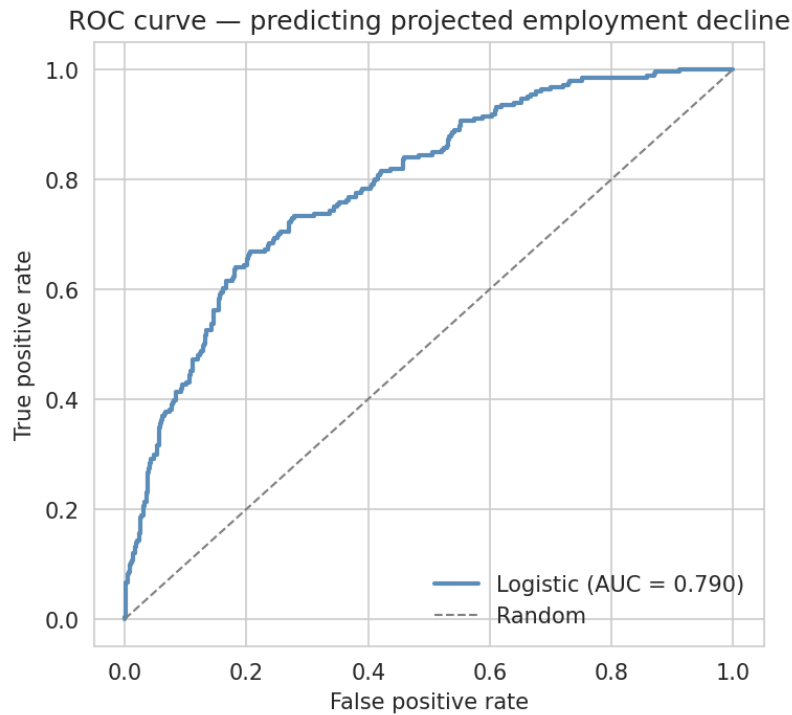
6. The relationship between Anthropic's AI risk tier and projected decline is non-monotonic (Figure 2). The Very high exposure tier has the highest decline rate (52%), consistent with the AI-displacement narrative. But the Very low tier, or mainly manual occupations, also shows elevated decline (37%, above the 29% baseline), reflecting pre-existing pressure from mechanization, offshoring, and demographic decline in

agriculture and manufacturing. The "moderate" tier (largely healthcare practitioners and social services) has the lowest decline rate.



a. Logistic Regression performance and threshold analysis

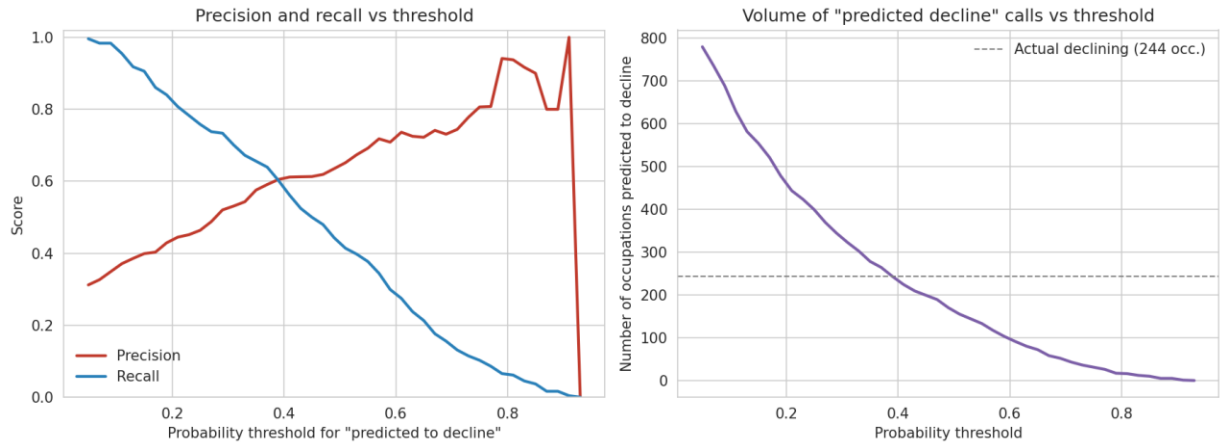
The base logistic model (8 features) achieves an in-sample AUC of 0.790 and a 5-fold cross-validated AUC of 0.773 ± 0.031 . The companion linear regression achieves CV R^2 of 0.177 ± 0.100 . The model is useful for ranking occupations by relative risk; the modest R^2 indicates substantial unexplained variance, and the model should not be treated as a precise forecast.



The choice of probability cut-off trades precision against recall. Table 1 shows performance across seven thresholds:

Threshold	Flagged occupations	Precision	Recall	F1
0.20	459	0.44	0.82	0.57
0.25	399	0.46	0.76	0.57
0.30	337	0.52	0.72	0.61
0.35	278	0.58	0.66	0.61
0.40	234	0.61	0.58	0.59
0.45	199	0.61	0.50	0.55
0.50	162	0.64	0.43	0.51

F1 peaks at threshold 0.30–0.35 (recall 66–72% at precision 52–58%). For workforce planning where missing a declining occupation is more costly than a false alarm, thresholds in the 0.25–0.35 range are appropriate; for more conservative flagging, 0.40–0.50 is better calibrated.

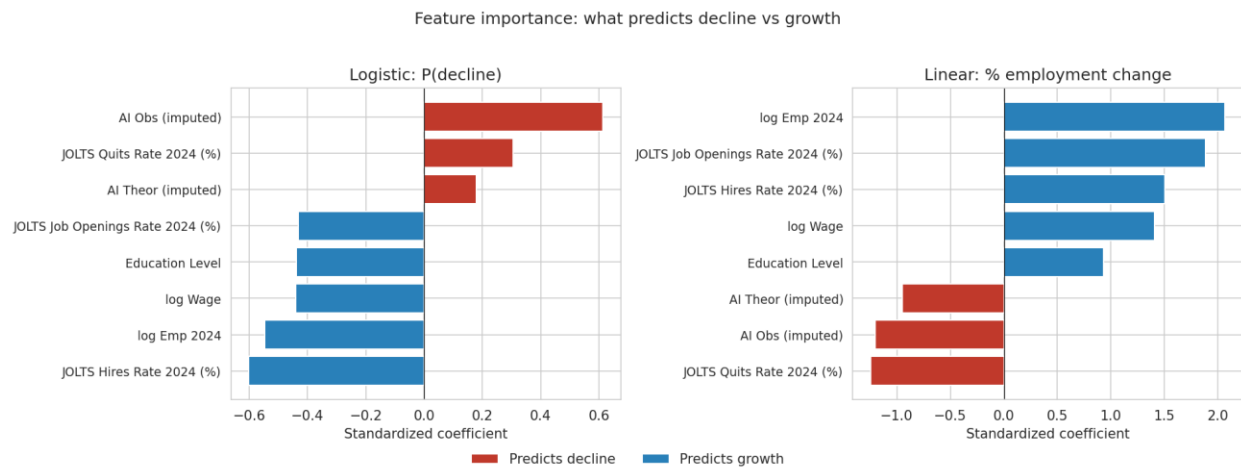


b. Identifying characteristics of employment decline

Table 2 shows standardized logistic coefficients and odds ratios for the eight base features. AI Observed Exposure has the largest positive (decline-predicting) coefficient (+0.61, odds ratio 1.85), meaning a one-standard-deviation increase in observed Claude usage nearly doubles the odds of projected decline. JOLTS Hires Rate has the largest negative (growth-predicting) coefficient (−0.60, odds ratio 0.55).

Feature	Standardized coef.	Odds ratio	Direction
AI Obs (imputed)	0.61	1.85	Predicts decline
JOLTS Hires Rate 2024 (%)	-0.60	0.55	Predicts growth
log Emp 2024	-0.55	0.58	Predicts growth
log Wage	-0.44	0.64	Predicts growth
Education Level	-0.44	0.64	Predicts growth
JOLTS Job Openings Rate 2024 (%)	-0.43	0.65	Predicts growth
JOLTS Quits Rate 2024 (%)	0.30	1.36	Predicts decline
AI Theor (imputed)	0.18	1.20	Predicts decline

The linear and logistic models agree on the direction of every feature (Figure 6), which is reassurance that the patterns are not artifacts of binarization. Red bars indicate predictors of decline in both panels; blue bars indicate predictors of growth.



c. Skill-level analysis

Univariate Pearson correlations between each of the 35 O*NET skills and the binary decline target (Figure 7) reveal a clear pattern. Skills associated with decline (positive correlation, in red) are dominated by manual/operational skills: Equipment Maintenance ($r = +0.12$), Repairing ($+0.11$), Operation and Control ($+0.09$), Operations Monitoring ($+0.07$). Skills protective against decline (negative correlation, in blue) are higher-order cognitive and managerial: Active Learning (-0.31), Critical Thinking (-0.30), Systems Evaluation (-0.30), Systems Analysis (-0.30), Personnel Management (-0.29).

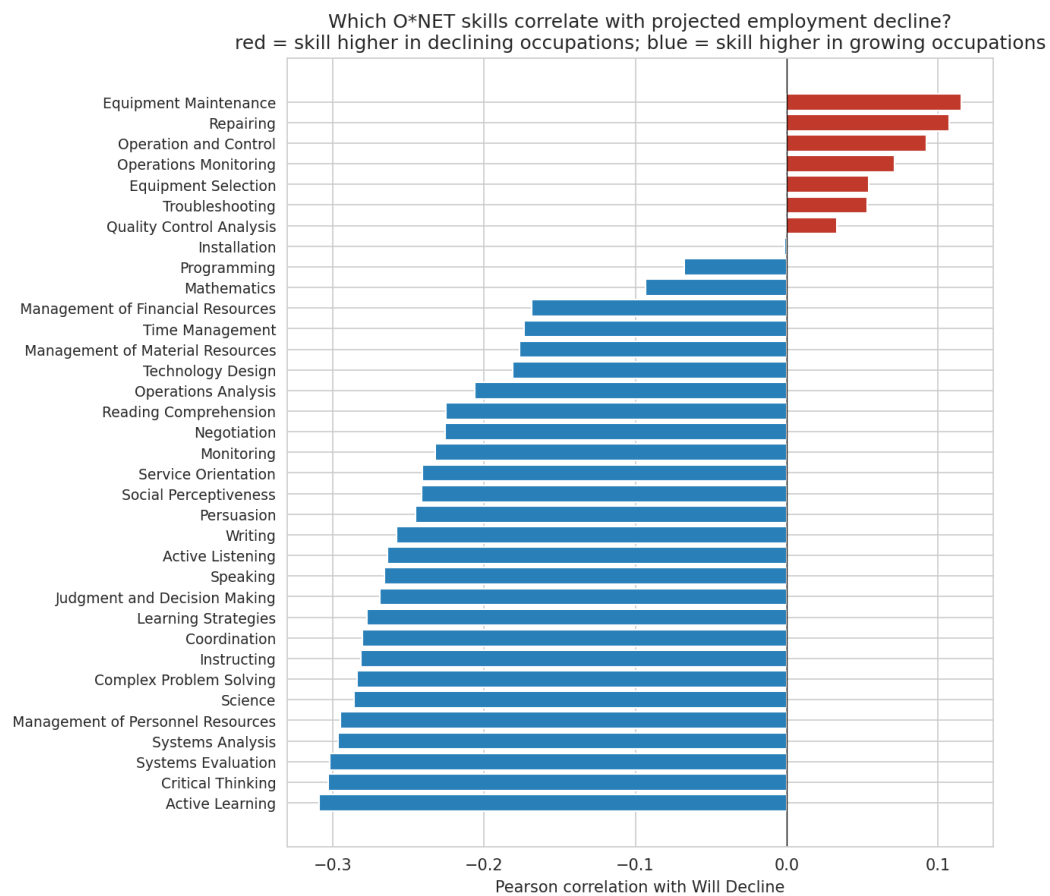


Figure 7. Correlation of each of the 35 O*NET skills with the binary decline target.

Plotting each skill's correlation with AI exposure against its correlation with decline (Figure 8) yields the report's most interesting visual finding: the "displacement quadrant" (top-right: high AI exposure AND associated with decline) is empty. The skills most associated with projected decline (top-left cluster) are the least AI-exposed. The skills most exposed to AI (Programming, Reading Comprehension, Speaking, Writing) are associated with employment growth rather than decline.

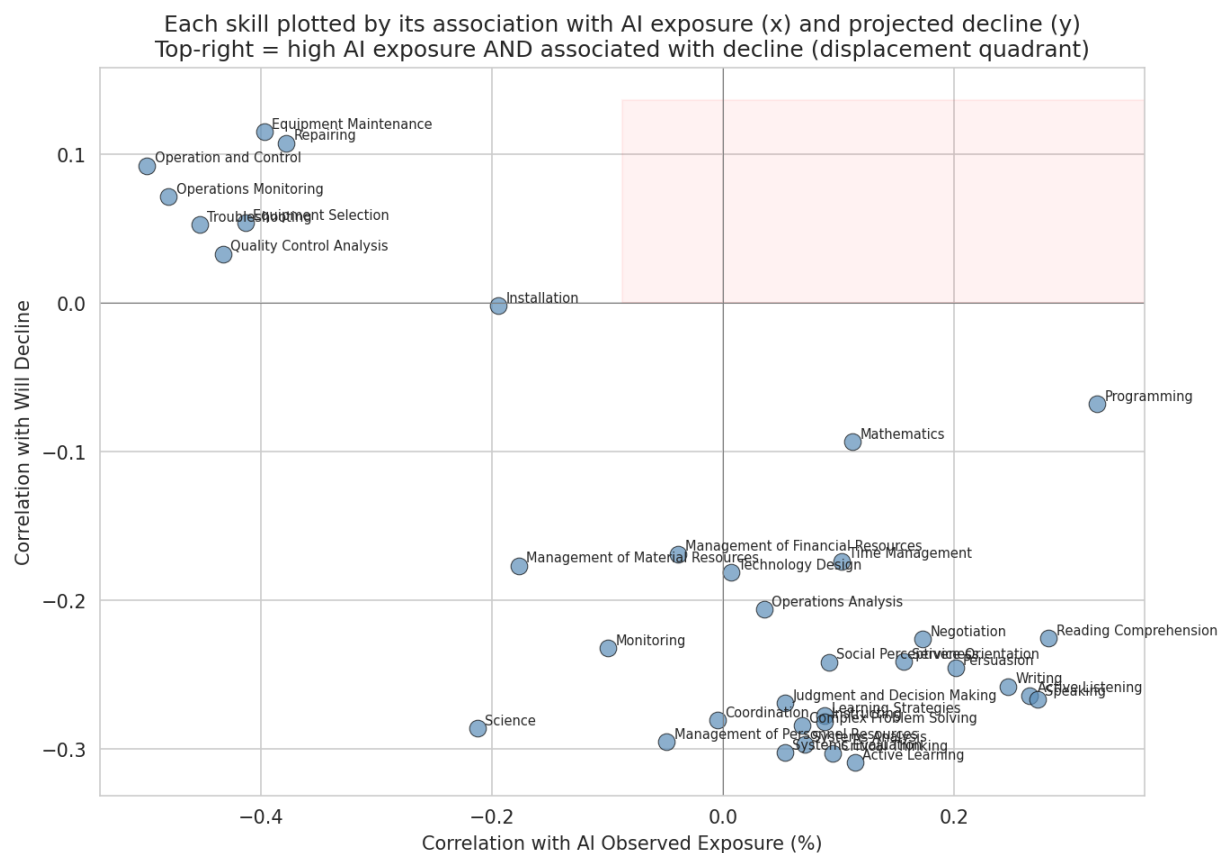


Figure 8. Quadrant chart: each skill plotted by its correlation with AI exposure (x) and with projected decline (y).

Principal Component Analysis on the 35 standardized skills extracts highly interpretable combined-skill factors. The first two components together explain 67% of the variance:

- PC1 (49% of variance) — a general cognitive/communication index loaded positively on Reading Comprehension, Writing, Speaking, Active Listening, Critical Thinking, Active Learning, Judgment and Decision Making, and Complex Problem Solving. Effectively a "knowledge-work" index.

- PC2 (18% of variance) — a physical/operational index loaded on Quality Control Analysis, Troubleshooting, Equipment Selection, Operations Monitoring, Repairing, and Equipment Maintenance. A "technical/blue-collar" index.
- PC3 (7%) — STEM (Programming, Mathematics, Science) versus people/service skills (Coordination, Personnel Management, Service Orientation, Social Perceptiveness).
- PC4 (5%) — financial/material resource management versus teaching/learning.

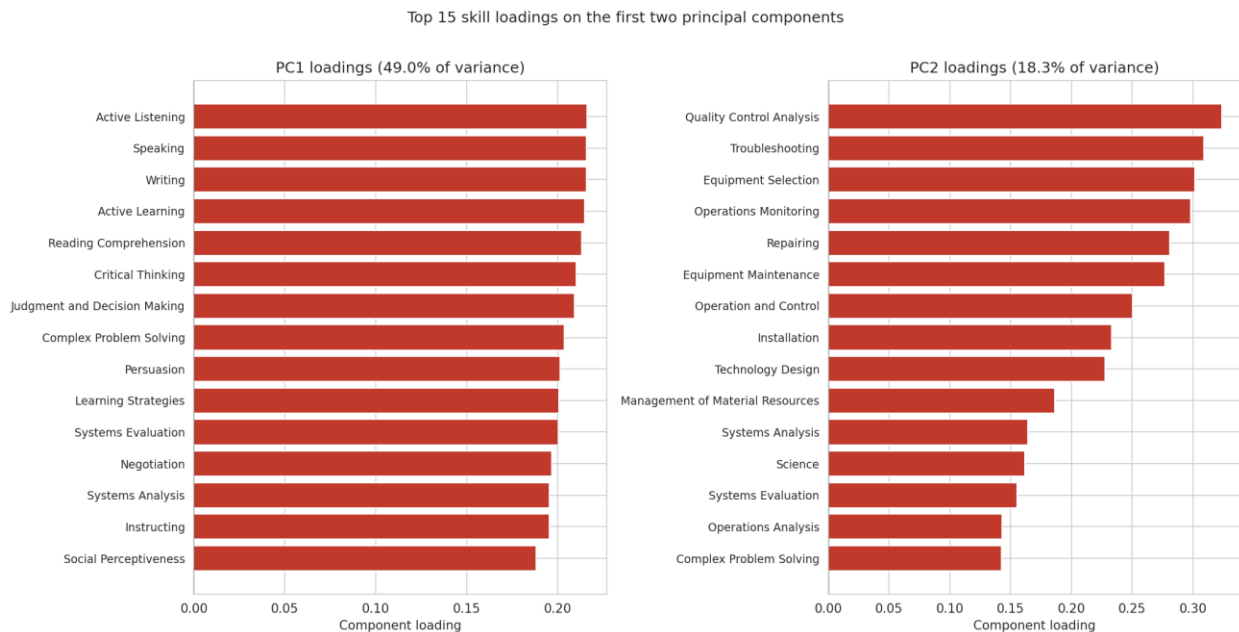


Figure 9. Top 15 skill loadings on the first two principal components.

d. Comparative Model (Skills, PCA, and no skills included)

On the 751-occupation skills subset, three models were compared (Table 3, Figure 10): base features only (8), base + 35 raw skills (43), and base + 4 PCs (12). Cross-validated AUC and R^2 are essentially identical across the three; error bars overlap entirely.

Model	Features	CV AUC	CV R^2
base	8	0.785 ± 0.025	+0.191 ± 0.074
base + skills	43	0.783 ± 0.043	+0.174 ± 0.108

Model	Features	CV AUC	CV R ²
base + PCs	12	0.783 ± 0.031	+0.188 ± 0.107

Table 3. Three-way model comparison on the 751-occupation subset with skills data.

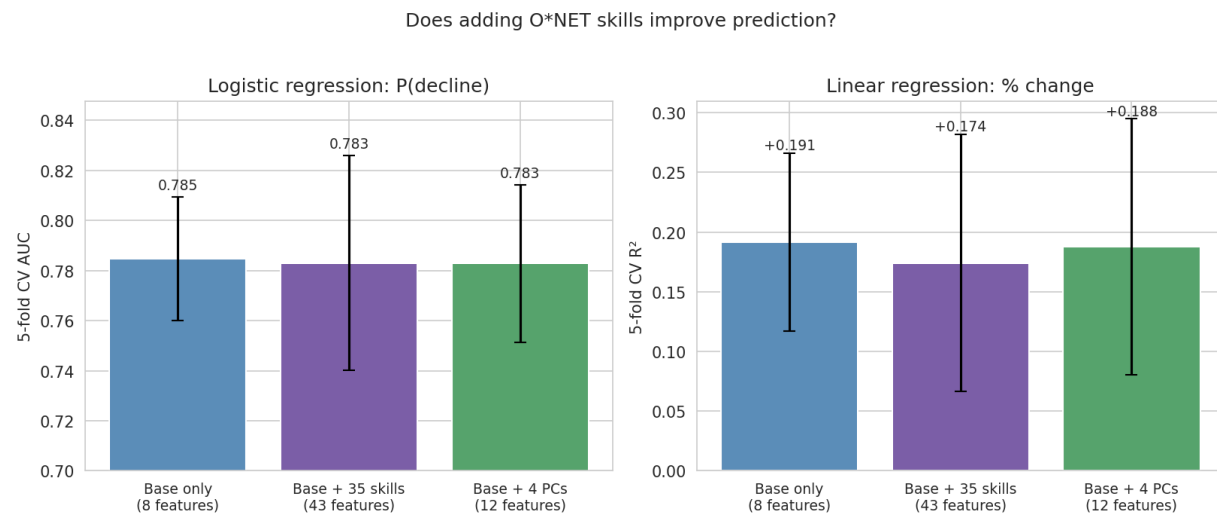


Figure 10. Cross-validated AUC (left) and R² (right) for the three model specifications.

In the base+PCs specification (Figure 11), only PC1 carries substantial weight (standardized coefficient -0.55 , predicting growth). PC2 through PC4 are near zero. The general cognitive-skills component is the only skill-derived feature that meaningfully adds to prediction, and it is largely captured by education level and log wage, which are themselves correlates of cognitive complexity.

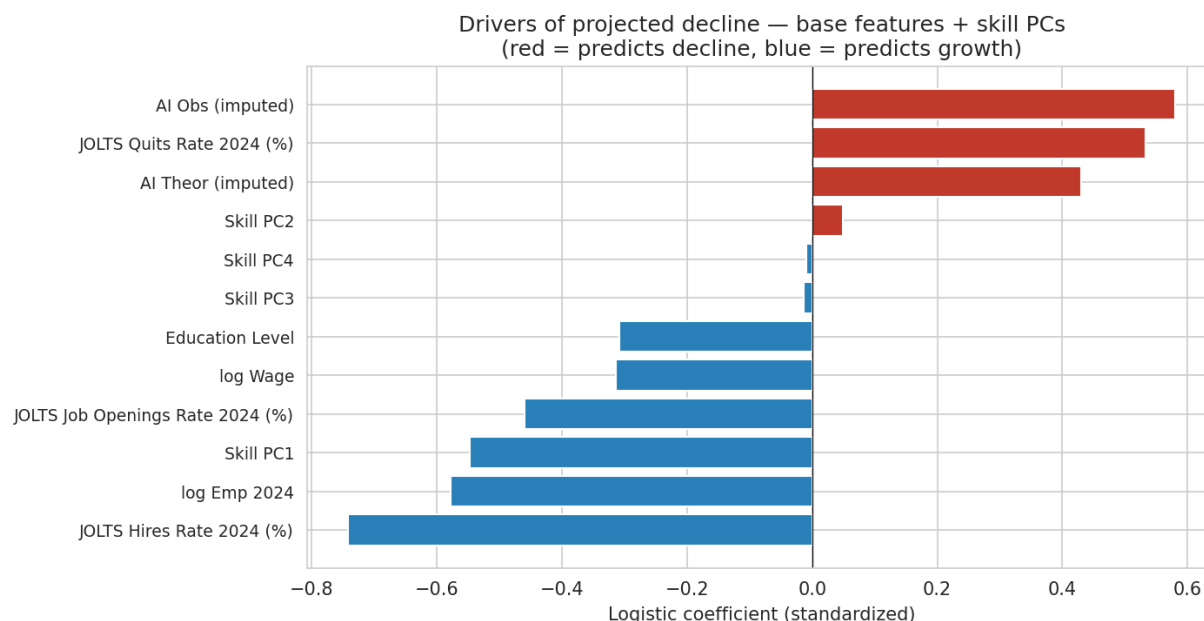


Figure 11. Logistic coefficients for the base + skill PCs model.

7. Discussion and Insights

Two findings stand out. First, projected employment decline through 2034 is associated with manual/operational skills (Equipment Maintenance, Repairing, Operation and Control) much more than with the skills currently being substituted by AI. The decline-associated skills are, in fact, the least AI-exposed of the 35 measured. This complicates simple narratives that frame BLS projections as AI-driven: the 2024–2034 decline patterns reflect long-running structural pressures, mechanization, offshoring, demographic shifts in agriculture, that pre-date generative AI by decades.

Second, adding O*NET skill features to the base model produces no measurable predictive lift. This is a substantively meaningful null result, not a methodological failure: it means the predictive signal in skills is already captured by features that proxy for skill mix indirectly (AI exposure, education level, wage, employment size). The skills nonetheless have substantial interpretive value. They let us name specific skills as protective or at-risk, which is much more actionable for career counseling and workforce planning than a single black-box probability.

Limitations are worth flagging. The model achieves moderate discriminative power (CV AUC $\approx$ 0.77–0.79); the linear model explains only about 18% of the variance in projected

percent change, leaving 82% unexplained by the available features. BLS projections themselves incorporate inputs not in this file — demographic forecasts, industry demand models, capital substitution assumptions, immigration policy — that materially drive the projections. JOLTS rates are industry averages rather than occupation-specific; the AI exposure data is mostly at the SOC major-group level rather than occupation-specific (the per-occupation file would tighten this). And all numbers are projections through 2034, not measured outcomes; the actual labor market path will depend on shocks and policy choices that we cannot observe.

Two extensions would meaningfully advance this work. First, ingesting the full per-occupation AI exposure file from the Anthropic Economic Index release (currently, the values are mostly at the SOC major-group level, with per-occupation overrides only for the ten most-exposed roles). Second, replacing the hand-built SOC-major-group-to-NAICS lookup for JOLTS with an OEWS-weighted crosswalk that uses where each occupation actually concentrates by industry. Both improvements would sharpen the inputs without changing the modeling framework.

8. Use of AI Tools

This project incorporates two distinct uses of AI. First, the Anthropic Economic Index (Massenkoff & McCrory, March 2026) is used as a data input: it provides theoretical and observed AI exposure measures by occupation, derived from analysis of actual Claude usage on tasks aligned to O*NET. The "observed exposure" measure in particular reflects empirical Claude usage in real workflows rather than expert speculation about what AI could do. In addition, the data model chart was generated in Claude.

9. Conclusion

A logistic regression on 832 U.S. occupations predicts projected employment decline by 2034 with a cross-validated AUC of 0.77, using eight features capturing AI exposure, education, wage, employment size, and industry churn. The model is useful for ranking occupations by relative risk but not for individual point forecasts. The strongest predictors of decline are high AI observed exposure, low industry hire rate, small occupation size, and lower entry-level education; the strongest predictors of growth are higher education, higher wages, and higher industry hires.

O*NET skill data, joined for 90% of occupations, adds no measurable predictive lift but provides crucial interpretive content. The skills most associated with projected decline are manual/operational (Equipment Maintenance, Repairing, Operation and Control); the skills most protective are higher-order cognitive (Active Learning, Critical Thinking, Systems Analysis). Crucially, the skills most exposed to AI (Programming, Reading Comprehension, Speaking, Writing) are not the skills associated with decline — they are associated with growth. The 2024–2034 BLS projections, on the evidence available here, are more consistent with continued structural decline in manual/operational work than with AI-driven contraction of cognitive work. This is a useful frame for the policy conversation: AI exposure and projected decline are partially orthogonal, and conflating them risks aiming workforce-transition resources at the wrong workers.

Sources:

- BLS Employment Projections, 2024–2034 release. <https://www.bls.gov/emp/>
- BLS JOLTS news release tables 16, 18, 22 (March 2026).
<https://www.bls.gov/news.release/jolts.toc.htm>
- Anthropic Economic Index — Massenkoff & McCrory, "Labor market impacts of AI" (March 2026). <https://www.anthropic.com/research/labor-market-impacts>
- O*NET database version 30.2, Skills file. <https://www.onetcenter.org/database.html>
- Eloundou, Manning, Mishkin & Rock (2023). GPTs are GPTs. arXiv:2303.10130.